

LEO ZHOU

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Education

Rice University

August 2025 – December 2028

Bachelor of Science in Computer Science

Houston, Texas

GPA: 3.92/4.00 **Relevant Coursework:** Data Structures & Algorithms, Linear Algebra, Computer Organization, Discrete Math, Intro to Computer Engineering, Digital Logic Design, Multivariable Calculus

Technical Skills

Languages: Go, Python, C/C++, Swift, JavaScript/TypeScript, SQL (PostgreSQL), HTML/CSS

Web & Backend: React, Node.js, FastAPI, Flask, REST APIs, WebSockets, Redis (Pub/Sub)

Cloud & Infrastructure: AWS (ECS, SQS), Terraform, Docker, GitHub Actions (CI/CD), Git, k6, Locust, Sentry

Machine Learning: PyTorch, ONNX, pybind11

Embedded: STM32 (bare-metal), FreeRTOS, ESP32, UART/SPI/I²C

Experience

CanSemi

June 2026 – Present

Software Engineer Intern

Guangzhou, China

- Built event-driven binary parser extracting parametric test records from 500+ ATE datalogs, used by 6 test engineers
- Designed normalized PostgreSQL schema (lot/wafer/die/result) & batch pipeline loading 40 lots (~12,000 die) in <2 min/lot
- Developed a FastAPI backend with 8 query endpoints serving wafer/lot-level test data, decreasing reporting time by ~75%
- Built an interactive React/Canvas wafer map rendering 300+ die with hover and drill-down, replacing static Excel heatmaps

Rice Wind Energy

October 2025 – April 2026

Firmware Engineer

Houston, Texas

- Developed ESP32 firmware in C/C++ for a 3-phase wind turbine controller with RPM, power, and pitch-control sensing
- Built a 6-state safety state machine with fault latching/recovery and a perturb-and-observe peak-power tracking algorithm

A-Lab@Rice

November 2025 – February 2026

Undergraduate Researcher

Houston, Texas

- Built COMSOL Multiphysics simulations for a solar thermal desalination project (STREED) to optimize model efficiency
- Executed parameter sweeps and analyzed outputs to improve model performance and support design alterations

Projects

BlunderNet Arena — Scalable Engine-Serving Platform | Go, AWS, Terraform, React June 2026 – July 2026

- Built a Go platform serving BlunderNet publicly: stateless API fleet behind an ALB, game state in Redis, 500+ games
- Decoupled inference onto an SQS-fed ONNX worker fleet autoscaled on queue depth, holding p99 move latency <250 ms
- Streamed moves over WebSockets with Redis pub/sub fan-out; killing an API task mid-game is invisible to players
- Defined the stack (VPC, ECS, ALB, RDS, SQS) in Terraform with apply/destroy deploy windows; AWS cost <\$5/mo
- Load-tested with k6 to 4,000+ req/s at zero errors; batched positions per worker poll for ~2× engine throughput

BlunderNet — Self-Training Chess Engine | PyTorch, C++, pybind11, GitHub Actions June 2026 – Present

- Built an AlphaZero-style engine (residual CNN, policy/value heads) with a self-play RL loop automated on GitHub Actions
- Implemented PUCT MCTS in Python and C++ (pybind11), batching leaf evaluations via virtual loss for a 1.8× speedup
- Automated fresh Lichess top-player game ingest with cursor-based pagination and game-ID dedup
- Tracked strength via held-out move prediction, a 1,200-puzzle tactics suite, and daily Elo vs. Stockfish

SeatLive — Concurrency-Safe Booking Engine | FastAPI, PostgreSQL, Redis, WebSockets May 2026 – June 2026

- Built a real-time booking engine preventing double-booked seats via atomic conditional UPDATES with Redis TTL holds
- Implemented idempotent confirmation via client-supplied keys, short-circuiting retries instead of double-processing
- Proved correctness with CI concurrency tests firing 50 simultaneous requests at one seat, ensuring exactly one succeeds
- Streamed live seat-map updates via WebSockets + Redis pub/sub; Locust load test served 4,459 requests with zero failures
- Containerized app, Postgres, and Redis with Docker Compose; initialized Sentry to capture unhandled exceptions

Leadership / Extracurricular

Junior National Fencing Team Captain + Coach & Referee

January 2022 – March 2025

- Captained NZ's junior national team to Oceania titles; competed at Junior Commonwealth Games & World Championships
- Earned 27 national first-place finishes; coached 100+ youth fencers and refereed hundreds of bouts across Oceania